

Applied decision problems

- Examples
- Decisions under missing information
- Recommendation systems

Decision and Bayesian decision problems

A **decision problem** involves three spaces

- a **state-space** $\mathcal{Q}$...states live here, $q \in \mathcal{Q}$
- a **data-space** $\mathcal{W}$...data live here, $w \in \mathcal{W}$
- an **action-space** $\mathcal{A}$...actions live here, $\alpha \in \mathcal{A}$

Our main goal is to obtain a **decision rule**

$$D : \mathcal{W} \mapsto \mathcal{A}$$

A Bayesian decision problem consists of

- a **Bayesian model**

$$q \sim Q$$
$$w|q \sim P_q$$

- a **loss function**

$$L : \mathcal{A} \times \mathcal{Q} \mapsto (-\infty, +\infty)$$

The **posterior expected loss** is

$$\rho(\alpha, w) = \underbrace{\mathbb{E}_{q|w} L(\alpha, q)}_{\text{posterior expectation}}$$

The **Bayes rule** is

$$D^*(w) = \operatorname{argmin}_{\alpha \in \mathcal{A}} \rho(\alpha, w)$$

Clinical patient management

A **suspected patient** shows ambiguous symptoms

Based on demographics and historical data, there is

- 20% chance our patient has **condition A**
- 15% chance our patient has **condition B**
- 65% chance our patient has **no condition**

...which we denote with C_A

...which we denote with C_B

...which we denote with $C_{\emptyset}$

On a different facility, the patient ran one of two diagnostics:

- A **costly test** with no side effects

...which we denote with D_1

This test identifies C_A with 90% success rate and C_B with only 50% success rate

- A **costless test** with severe side effects

...which we denote with D_2

This test identifies both C_A and C_B with 85% success rate

We **lost record** of which diagnostic the patient ran, but we maintain the diagnostic's outcome

Our goal is to come up with a **treatment plan**

Clinical patient management

We form and solve a Bayesian decision problem

- Our problem's states are

...patient's actual condition

$$c \in \{C_A, C_B, C_\emptyset\}$$

- Our problem's data are

...diagnostic's outcome

$$w \in \{A, B, \emptyset\}$$

- Our problem's actions are

...treatment plans

$$t \in \{T_A, T_B, T_\emptyset\}$$

Clinical patient management

Our Bayesian model consists of

$$c \sim \text{Categorical}_{C_A, C_B, C_\emptyset} (0.20, 0.15, 0.65)$$

$$d \sim \text{Categorical}_{D_1, D_2} (\psi_{D_1}, \psi_{D_2})$$

$$w|c, d \sim \text{Categorical}_{A, B, \emptyset} (f_d(c))$$

- To model our diagnostics, we use

$$f_{D_1}(C_A) = (0.90, 0.00, 0.10)$$

$$f_{D_1}(C_B) = (0.00, 0.50, 0.50)$$

$$f_{D_1}(C_\emptyset) = (0.00, 0.00, 1.00)$$

$$f_{D_2}(C_A) = (0.85, 0.00, 0.15)$$

$$f_{D_2}(C_B) = (0.00, 0.85, 0.15)$$

$$f_{D_2}(C_\emptyset) = (0.00, 0.00, 1.00)$$

Clinical patient management

Our **Bayesian model** consists of

$$c \sim \text{Categorical}_{C_A, C_B, C_\emptyset} (0.20, 0.15, 0.65)$$

$$d \sim \text{Categorical}_{D_1, D_2} (\psi_{D_1}, \psi_{D_2})$$

$$w|c, d \sim \text{Categorical}_{A, B, \emptyset} (f_d(c))$$

- Our **posterior** is

$$\begin{aligned} p(c|w) &= \sum_d p(c, d|w) \propto \sum_d p(w|c, d)p(c, d) \\ &= \sum_d p(w|c, d)p(c)p(d) \\ &= p(c) \sum_d p(w|c, d)p(d) \\ &= p(c) (p(w|c, d = D_1)p(d = D_1) + p(w|c, d = D_2)p(d = D_2)) \\ &= \text{Cat}(c; \dots) (\text{Cat}(w; f_{D_1}(c))\psi_{D_1} + \text{Cat}(w; f_{D_2}(c))\psi_{D_2}) \\ &= \text{Cat}(c; \dots) \text{Cat}(w; \psi_{D_1} f_{D_1}(c) + \psi_{D_2} f_{D_2}(c)) = \text{Cat}(c; \dots) \end{aligned}$$

Clinical patient management

Our **loss** function is

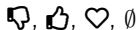
$$L(t, c) = \begin{cases} 0, & c = C_A, t = T_A \\ 0.8, & c = C_B, t = T_A \\ 1.0, & c = C_\emptyset, t = T_A \\ 0.6, & c = C_A, t = T_B \\ 0, & c = C_B, t = T_B \\ 0.9, & c = C_\emptyset, t = T_B \\ 0.3, & c = C_A, t = T_\emptyset \\ 0.4, & c = C_B, t = T_\emptyset \\ 0, & c = C_\emptyset, t = T_\emptyset \end{cases}$$

A recommendation system

We own rights to a **movie database** with five genres ...we ignore multi-genre movies and movie attributes

'horror', 'thriller', 'action', 'drama', 'comedy'

Our platform streams movies and **records reactions** ...we ignore time-stamps and viewer attributes



We aim to develop an **automated recommendation algorithm**

...good recommendations will make viewers happy and maintain our subscription

...bad recommendations will make viewers unhappy and leave our subscription

We will form and solve a **decision problem**

A recommendation system

We consider a **single viewer**. Our viewer has engaged in **past viewing rounds** which we index with

$$n = 1, \dots, N$$

- Our **problem's states** are the viewer's actual preferences ...encoded as a vector in the probability simplex

$$\pi = (\pi_{\text{'horror'}}, \pi_{\text{'thriller'}}, \pi_{\text{'action'}}, \pi_{\text{'drama'}}, \pi_{\text{'comedy'}}) \in \mathbb{R}_{\text{prob}}^5$$

- Our **problem's data** are the viewer's past movies and reactions ...encoded as two sequences

$$\mu_n \in \{\text{'horror'}, \text{'thriller'}, \text{'action'}, \text{'drama'}, \text{'comedy'}\}$$

$$\omega_n \in \{\text{👎}, \text{👍}, \text{❤️}, \emptyset\}$$

- Our **problem's actions** are the recommendations

$$\alpha \in \{\text{'horror'}, \text{'thriller'}, \text{'action'}, \text{'drama'}, \text{'comedy'}\}$$

for the $(N + 1)^{\text{th}}$ viewing round

A recommendation system

Our **Bayesian model** consists of

$$\pi \sim \text{Dirichlet}_{\text{'horror'}, \text{'thriller'}, \text{'action'}, \text{'drama'}, \text{'comedy'}} (\gamma_{\text{'horror'}}, \gamma_{\text{'thriller'}}, \gamma_{\text{'action'}}, \gamma_{\text{'drama'}}, \gamma_{\text{'comedy'}})$$

$$\mu_n | \pi \sim \text{Categorical}_{\text{'horror'}, \text{'thriller'}, \text{'action'}, \text{'drama'}, \text{'comedy'}} (\pi), \quad n = 1, \dots, N$$

$$\omega_n | \mu_n, \pi \sim \text{Categorical}_{\text{👍, 👎, ❤️, } \emptyset} \left(f_{\text{👍}}^{\mu_n}(\pi), f_{\text{👎}}^{\mu_n}(\pi), f_{\text{❤️}}^{\mu_n}(\pi), f_{\emptyset}^{\mu_n}(\pi) \right), \quad n = 1, \dots, N$$

- To model **non-informative** priors, we use

$$\gamma_{\text{'horror'}} = \gamma_{\text{'thriller'}} = \gamma_{\text{'action'}} = \gamma_{\text{'drama'}} = \gamma_{\text{'comedy'}} = 1$$

- To model **preferential** reactions, we use

$$f_{\text{👍}}^{\mu}(\pi) = \frac{6}{10} (1 - \pi_{\mu}) + \frac{1}{10}$$

$$f_{\text{👎}}^{\mu}(\pi) = \frac{5}{10} \pi_{\mu} + \frac{1}{10}$$

$$f_{\text{❤️}}^{\mu}(\pi) = \frac{1}{10} \pi_{\mu}$$

$$f_{\emptyset}^{\mu_n}(\pi) = \frac{2}{10}$$

A recommendation system

Our **loss** function is

$$L(\alpha, \pi) = 1 - \pi_\alpha$$

this way:

- good recommendations agree with our viewer's preference ...high preference π_α and low loss $L(\alpha, \pi)$
- bad recommendations disagree with our viewer's preference ...low preference π_α and high loss $L(\alpha, \pi)$

Our loss assumes all genres are **equally valuable/costful** to our platform

Alternative losses

$$L(\alpha, \pi) = \left(\max_{\alpha'} R_{\alpha'} \right) - R_\alpha \pi_\alpha$$

Assumes genres have **unequal rewards/costs**

$$L(\alpha, \pi) = \left(\left(\max_{\alpha'} R_{\alpha'} \right) - R_\alpha \pi_\alpha \right)^2$$

Avoids **over-saturation**

A recommendation system

- Ioannis engages with $N = 11$ movies

Our **posterior** is given by

$$\begin{aligned}
 p(\pi | \mu_{1:N}, \omega_{1:N}) &\propto p(\mu_{1:N}, \omega_{1:N} | \pi) p(\pi) \\
 &= p(\omega_{1:N} | \mu_{1:N}, \pi) \\
 &\quad \times p(\mu_{1:N} | \pi) p(\pi) \\
 &= \left[\prod_{n=1}^N p(\omega_n | \mu_{1:N}, \pi) \right] \\
 &\quad \times \left[\prod_{n=1}^N p(\mu_n | \pi) \right] p(\pi) \\
 &= \left[\prod_{n=1}^N \text{Cat}(\omega_n; f^{\mu_n}(\pi)) \right] \\
 &\quad \times \underbrace{\left[\prod_{n=1}^N \text{Cat}(\mu_n; \pi) \right]}_{\propto \text{Dir}(\pi; \gamma')} \text{Dir}(\pi; \gamma)
 \end{aligned}$$

n	μ_n	ω_n	
1	'comedy'		<i>Blues Brothers 2000</i>
2	'thriller'		<i>The Fugitive</i>
3	'horror'		<i>The Conjuring</i>
4	'drama'		<i>Good Will Hunting</i>
5	'action'		<i>Die Hard with a Vengeance</i>
6	'horror'		<i>Halloween: Resurrection</i>
7	'thriller'		<i>Se7en</i>
8	'drama'		<i>The Departed</i>
9	'horror'		<i>Bride of Chucky</i>
10	'thriller'		<i>The Manchurian Candidate</i>
11	'action'		<i>John Wick</i>

A recommendation system

- Ioannis engages with $N = 11$ movies

Our **posterior** is given by

$$p(\pi | \mu_{1:N}, \omega_{1:N}) \propto \left[\prod_{n=1}^N \text{Cat}(\omega_n; f^{\mu_n}(\pi)) \right] \text{Dir}(\pi; \gamma')$$

$$\approx \text{Dir}(\pi; \gamma'')$$

With posterior **weights** given by

$$\begin{aligned} \gamma'_{\text{'horror'}} &= \gamma_{\text{'horror'}} + 3 & \gamma''_{\text{'horror'}} &= \gamma'_{\text{'horror'}} + 4.7 \\ \gamma'_{\text{'thriller'}} &= \gamma_{\text{'thriller'}} + 3 & \gamma''_{\text{'thriller'}} &= \gamma'_{\text{'thriller'}} + 2.3 \\ \gamma'_{\text{'action'}} &= \gamma_{\text{'action'}} + 2 & \gamma''_{\text{'action'}} &= \gamma'_{\text{'action'}} - 0.9 \\ \gamma'_{\text{'drama'}} &= \gamma_{\text{'drama'}} + 2 & \gamma''_{\text{'drama'}} &= \gamma'_{\text{'drama'}} - 0.8 \\ \gamma'_{\text{'comedy'}} &= \gamma_{\text{'comedy'}} + 1 & \gamma''_{\text{'comedy'}} &= \gamma'_{\text{'comedy'}} - 1.1 \end{aligned}$$

n	μ_n	ω_n	
1	'comedy'		<i>Blues Brothers 2000</i>
2	'thriller'		<i>The Fugitive</i>
3	'horror'		<i>The Conjuring</i>
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A recommendation system

- Ioannis engages with $N = 11$ movies

Our **posterior** is given by

$$p(\pi | \mu_{1:N}, \omega_{1:N}) \approx \text{Dir}(\pi; \gamma'')$$

With posterior **weights** given by

$$\gamma''_{\text{'horror'}} = 8.7$$

$$\gamma''_{\text{'thriller'}} = 6.3$$

$$\gamma''_{\text{'action'}} = 2.1$$

$$\gamma''_{\text{'drama'}} = 2.2$$

$$\gamma''_{\text{'comedy'}} = 0.9$$

Our **expected posterior loss** is

$$\rho(\alpha) \approx \int d\pi (1 - \pi_\alpha) \text{Dir}(\pi; \gamma'')$$

$$= 1 - \int d\pi \pi_\alpha \text{Dir}(\pi; \gamma'') = 1 - \frac{\gamma''_\alpha}{\sum_{\alpha'} \gamma''_{\alpha'}}$$

n	μ_n	ω_n	
1	'comedy'		<i>Blues Brothers 2000</i>
2	'thriller'		<i>The Fugitive</i>
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A recommendation system

- Ioannis engages with $N = 11$ movies

Our **posterior** is given by

$$p(\pi | \mu_{1:N}, \omega_{1:N}) \approx \text{Dir}(\pi; \gamma'')$$

Our **expected posterior losses** are

$$\rho(\text{'horror'}) \approx 0.57$$

$$\rho(\text{'thriller'}) \approx 0.69$$

$$\rho(\text{'action'}) \approx 0.90$$

$$\rho(\text{'drama'}) \approx 0.89$$

$$\rho(\text{'comedy'}) \approx 0.96$$

Our **Bayes recommendation** is

$$\alpha^* = \text{'horror'}$$

n	μ_n	ω_n	
1	'comedy'		<i>Blues Brothers 2000</i>
2	'thriller'	$\emptyset$	<i>The Fugitive</i>
3	'horror'		<i>The Conjuring</i>
4	'drama'		<i>Good Will Hunting</i>
5	'action'	$\emptyset$	<i>Die Hard with a Vengeance</i>
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